

Then the work done by $\vec{F}$ is defined to be

$$W = \|\vec{F}_{\text{parallel}}\| \|\vec{d}\|.$$

We see from Figure 13.33 that $\vec{F}_{\text{parallel}}$ has magnitude $\|\vec{F}\| \cos \theta$. So the work is given by the dot product:

$$W = (\|\vec{F}\| \cos \theta) \|\vec{d}\| = \|\vec{F}\| \|\vec{d}\| \cos \theta = \vec{F} \cdot \vec{d}.$$

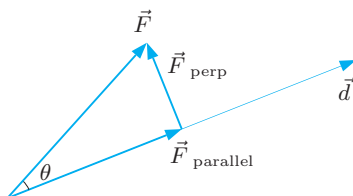


Figure 13.33: Resolving the force $\vec{F}$ into two forces, one parallel to $\vec{d}$, one perpendicular to $\vec{d}$

The formula $W = \vec{F} \cdot \vec{d}$ holds when $\pi/2 < \theta \leq \pi$ also. In that case, the work done by the force is negative and the object is moving against the force. Thus, we have the following definition:

The **work**, W , done by a force $\vec{F}$ acting on an object through a displacement $\vec{d}$ is given by

$$W = \vec{F} \cdot \vec{d}.$$

Example 9

How much work does the wind do on the sailboat from Example 8 if the boat moves 20 m and the wind's force is 120 newtons?

Solution

From Example 8, we know that the force of the wind $\vec{F}$ makes a 30° angle with the boat's displacement $\vec{d}$. Since $\|\vec{F}\| = 120$ and $\|\vec{d}\| = 20$, the work done by the wind on the boat is

$$W = \vec{F} \cdot \vec{d} = \|\vec{F}\| \|\vec{d}\| \cos 30^\circ = 2078.461 \text{ joules.}$$

Notice that if the vectors $\vec{F}$ and $\vec{d}$ are parallel and in the same direction, with magnitudes F and d , then $\cos \theta = \cos 0 = 1$, so $W = \|\vec{F}\| \|\vec{d}\| = Fd$, which is the original definition. When the vectors are perpendicular, $\cos \theta = \cos(\pi/2) = 0$, so $W = 0$ and no work is done in the technical definition of the word. For example, if you carry a heavy box across the room at the same horizontal height, no work is done by gravity because the force of gravity is vertical but the motion is horizontal.

Exercises and Problems for Section 13.3

Exercises

For Exercises 1–9, perform the following operations on the given 3-dimensional vectors.

$$\vec{a} = 2\vec{j} + \vec{k} \quad \vec{b} = -3\vec{i} + 5\vec{j} + 4\vec{k} \quad \vec{c} = \vec{i} + 6\vec{j}$$

$$\vec{y} = 4\vec{i} - 7\vec{j} \quad \vec{z} = \vec{i} - 3\vec{j} - \vec{k}$$

1. $\vec{a} \cdot \vec{y}$

2. $\vec{c} \cdot \vec{y}$

3. $\vec{a} \cdot \vec{b}$

5. $\vec{c} \cdot \vec{a} + \vec{a} \cdot \vec{y}$

7. $(\vec{a} \cdot \vec{b})\vec{a}$

9. $((\vec{c} \cdot \vec{c})\vec{a}) \cdot \vec{a}$

4. $\vec{a} \cdot \vec{z}$

6. $\vec{a} \cdot (\vec{c} + \vec{y})$

8. $(\vec{a} \cdot \vec{y})(\vec{c} \cdot \vec{z})$

In Exercises 10–14, find a normal vector to the plane.

10. $2x + y - z = 5$
11. $2(x - z) = 3(x + y)$
12. $1.5x + 3.2y + z = 0$
13. $z = 3x + 4y - 7$
14. $\pi(x - 1) = (1 - \pi)(y - z) + \pi$

In Exercises 15–21, find an equation of a plane that satisfies the given conditions.

15. Through $(1, 5, 2)$ perpendicular to $3\vec{i} - \vec{j} + 4\vec{k}$
16. Through $(2, -1, 3)$ perpendicular to $5\vec{i} + 4\vec{j} - \vec{k}$.
17. Through $(1, 3, 5)$ and normal to $\vec{i} - \vec{j} + \vec{k}$.

18. Perpendicular to $5\vec{i} + \vec{j} - 2\vec{k}$ and passing through $(0, 1, -1)$.
19. Parallel to $2x + 4y - 3z = 1$ and through $(1, 0, -1)$.
20. Through $(-2, 3, 2)$ and parallel to $3x + y + z = 4$.
21. Perpendicular to $\vec{v} = 2\vec{i} - 3\vec{j} + 5\vec{k}$ and through $(4, 5, -2)$.

In Exercises 22–26, compute the angle between the vectors.

22. $\vec{i} + \vec{j} + \vec{k}$ and $\vec{i} - \vec{j} - \vec{k}$.
23. $\vec{i} + \vec{k}$ and $\vec{j} - \vec{k}$.
24. $\vec{i} + \vec{j} - \vec{k}$ and $2\vec{i} + 3\vec{j} + \vec{k}$.
25. $\vec{i} + \vec{j}$ and $\vec{i} + 2\vec{j} - \vec{k}$.
26. $\vec{i}$ and $2\vec{i} + 3\vec{j} - \vec{k}$.

Problems

27. Give a unit vector

- (a) In the same direction as $\vec{v} = 2\vec{i} + 3\vec{j}$.
- (b) Perpendicular to $\vec{v}$.

28. A plane has equation $z = 5x - 2y + 7$.

- (a) Find a value of λ making the vector $\lambda\vec{i} + \vec{j} + 0.5\vec{k}$ normal to the plane.
- (b) Find a value of a so that the point $(a + 1, a, a - 1)$ lies on the plane.

29. Consider the plane $5x - y + 7z = 21$.

- (a) Find a point on the x -axis on this plane.
- (b) Find two other points on the plane.
- (c) Find a vector perpendicular to the plane.
- (d) Find a vector parallel to the plane.

30. (a) Find a vector perpendicular to the plane $z = 2 + 3x - y$.

- (b) Find a vector parallel to the plane.

31. (a) Find a vector perpendicular to the plane $z = 2x + 3y$.

- (b) Find a vector parallel to the plane.

32. Match the planes in (a)–(d) with one or more of the descriptions in (I)–(IV). No reasons needed.

- (a) $3x - y + z = 0$
- (b) $4x + y + 2z - 5 = 0$
- (c) $x + y = 5$
- (d) $x = 5$

- I Goes through the origin.
- II Has a normal vector parallel to the xy -plane.
- III Goes through the point $(0, 5, 0)$.
- IV Has a normal vector whose dot products with $\vec{i}$, $\vec{j}$, $\vec{k}$ are all positive.

33. Which pairs (if any) of vectors from the following list

- (a) Are perpendicular?
- (b) Are parallel?
- (c) Have an angle less than $\pi/2$ between them?

- (d) Have an angle of more than $\pi/2$ between them?

$$\vec{a} = \vec{i} - 3\vec{j} - \vec{k}, \quad \vec{b} = \vec{i} + \vec{j} + 2\vec{k},$$

$$\vec{c} = -2\vec{i} - \vec{j} + \vec{k}, \quad \vec{d} = -\vec{i} - \vec{j} + \vec{k}.$$

34. List any vectors that are parallel to each other and any vectors that are perpendicular to each other:

$$\vec{v}_1 = \vec{i} - 2\vec{j} \quad \vec{v}_2 = 2\vec{i} + 4\vec{j}$$

$$\vec{v}_3 = 3\vec{i} + 1.5\vec{j} \quad \vec{v}_4 = -1.2\vec{i} + 2.4\vec{j}$$

$$\vec{v}_5 = -5\vec{i} - 2.5\vec{j} \quad \vec{v}_6 = 12\vec{i} - 12\vec{j}$$

$$\vec{v}_7 = 4\vec{i} + 2\vec{j} \quad \vec{v}_8 = 3\vec{i} - 6\vec{j}$$

$$\vec{v}_9 = 0.70\vec{i} - 0.35\vec{j}$$

35. (a) Give a vector that is parallel to, but not equal to, $\vec{v} = 4\vec{i} + 3\vec{j}$.

- (b) Give a vector that is perpendicular to $\vec{v}$.

36. For what values of t are $\vec{u} = t\vec{i} - \vec{j} + \vec{k}$ and $\vec{v} = t\vec{i} + t\vec{j} - 2\vec{k}$ perpendicular? Are there values of t for which $\vec{u}$ and $\vec{v}$ are parallel?

37. Let θ be the angle between $\vec{v}$ and $\vec{w}$, with $0 < \theta < \pi/2$. What is the effect on $\vec{v} \cdot \vec{w}$ of increasing each of the following quantities? Does $\vec{v} \cdot \vec{w}$ increase or decrease?

- (a) $\|\vec{v}\|$
- (b) θ

38. Write $\vec{a} = 3\vec{i} + 2\vec{j} - 6\vec{k}$ as the sum of two vectors, one parallel, and one perpendicular, to $\vec{d} = 2\vec{i} - 4\vec{j} + \vec{k}$.

39. Find angle BAC if $A = (2, 2, 2)$, $B = (4, 2, 1)$, and $C = (2, 3, 1)$.

40. The points $(5, 0, 0)$, $(0, -3, 0)$, and $(0, 0, 2)$ form a triangle. Find the lengths of the sides of the triangle and each of its angles.

41. Let S be the triangle with vertices $A = (2, 2, 2)$, $B = (4, 2, 1)$, and $C = (2, 3, 1)$.

- (a) Find the length of the shortest side of S .
 (b) Find the cosine of the angle BAC at vertex A .

In Problems 42–47, given $\vec{v} = 3\vec{i} + 4\vec{j}$ and force vector $\vec{F}$, find:

- (a) The component of $\vec{F}$ parallel to $\vec{v}$.
 (b) The component of $\vec{F}$ perpendicular to $\vec{v}$.
 (c) The work, W , done by force $\vec{F}$ through displacement $\vec{v}$.

42. $\vec{F} = 4\vec{i} + \vec{j}$

43. $\vec{F} = 0.2\vec{i} - 0.5\vec{j}$

44. $\vec{F} = 9\vec{i} + 12\vec{j}$

45. $\vec{F} = -0.4\vec{i} + 0.3\vec{j}$

46. $\vec{F} = -3\vec{i} - 5\vec{j}$

47. $\vec{F} = -6\vec{i} - 8\vec{j}$

In Problems 48–51, the force on an object is $\vec{F} = -20\vec{j}$. For vector $\vec{v}$, find:

- (a) The component of $\vec{F}$ parallel to $\vec{v}$.
 (b) The component of $\vec{F}$ perpendicular to $\vec{v}$.
 (c) The work, W , done by force $\vec{F}$ through displacement $\vec{v}$.

48. $\vec{v} = 2\vec{i} + 3\vec{j}$

49. $\vec{v} = 5\vec{i} - \vec{j}$

50. $\vec{v} = 3\vec{j}$

51. $\vec{v} = 5\vec{i}$

52. A basketball gymnasium is 25 meters high, 80 meters wide and 200 meters long. For a half-time stunt, the cheerleaders want to run two strings, one from each of the two corners above one basket to the diagonally opposite corners of the gym floor. What is the cosine of the angle made by the strings as they cross?

53. A 100-meter dash is run on a track in the direction of the vector $\vec{v} = 2\vec{i} + 6\vec{j}$. The wind velocity $\vec{w}$ is $5\vec{i} + \vec{j}$ km/hr. The rules say that a legal wind speed measured in the direction of the dash must not exceed 5 km/hr. Will the race results be disqualified due to an illegal wind? Justify your answer.

54. An airplane is flying toward the southeast. Which of the following wind velocity vectors increases the plane's speed the most? Which slows down the plane the most?

$$\vec{w}_1 = -4\vec{i} - \vec{j} \quad \vec{w}_2 = \vec{i} - 2\vec{j} \quad \vec{w}_3 = -\vec{i} + 8\vec{j}$$

$$\vec{w}_4 = 10\vec{i} + 2\vec{j} \quad \vec{w}_5 = 5\vec{i} - 2\vec{j}$$

55. A canoe is moving with velocity $\vec{v} = 5\vec{i} + 3\vec{j}$ m/sec relative to the water. The velocity of the current in the water is $\vec{c} = \vec{i} + 2\vec{j}$ m/sec.

- (a) What is the speed of the current?
 (b) What is the speed of the current in the direction of the canoe's motion?

56. Find a vector that bisects the smaller of the two angles formed by $3\vec{i} + 4\vec{j}$ and $5\vec{i} - 12\vec{j}$.

57. Find the shortest distance between the planes $2x - 5y + z = 10$ and $z = 5y - 2x$.

58. A street vendor sells six items, with prices p_1 dollars per unit, p_2 dollars per unit, and so on. The vendor's price vector is $\vec{p} = (p_1, p_2, p_3, p_4, p_5, p_6) = (1.00, 3.50, 4.00, 2.75, 5.00, 3.00)$. The vendor sells q_1 units of the first item, q_2 units of the second item, and so on. The vendor's quantity vector is $\vec{q} = (q_1, q_2, q_3, q_4, q_5, q_6) = (43, 57, 12, 78, 20, 35)$. Find $\vec{p} \cdot \vec{q}$, give its units, and explain its significance to the vendor.

59. A course has four exams, weighted 10%, 15%, 25%, 50%, respectively. The class average on each of these exams is 75%, 91%, 84%, 87%, respectively. What do the vectors $\vec{a} = (0.75, 0.91, 0.84, 0.87)$ and $\vec{w} = (0.1, 0.15, 0.25, 0.5)$ represent, in terms of the course? Calculate the dot product $\vec{w} \cdot \vec{a}$. What does it represent, in terms of the course?

60. A consumption vector of three goods is defined by $\vec{x} = (x_1, x_2, x_3)$, where x_1 , x_2 and x_3 are the quantities consumed of the three goods. A budget constraint is represented by the equation $\vec{p} \cdot \vec{x} = k$, where $\vec{p}$ is the price vector of the three goods and k is a constant. Show that the difference between two consumption vectors corresponding to points satisfying the same budget constraint is perpendicular to the price vector $\vec{p}$.

61. What does Property 2 of the dot product in the box on page 735 say geometrically?

62. Show that the vectors $(\vec{b} \cdot \vec{c})\vec{a} - (\vec{a} \cdot \vec{c})\vec{b}$ and $\vec{c}$ are perpendicular.

63. Show why each of the properties of the dot product in the box on page 735 follows from the algebraic definition of the dot product:

$$\vec{v} \cdot \vec{w} = v_1 w_1 + v_2 w_2 + v_3 w_3.$$

64. Show that if $\vec{u}$ and $\vec{v}$ are two vectors such that

$$\vec{u} \cdot \vec{w} = \vec{v} \cdot \vec{w}$$

for every vector $\vec{w}$, then

$$\vec{u} = \vec{v}.$$

65. Show that

$$\frac{\vec{u}}{\|\vec{u}\|^2} - \frac{\vec{v}}{\|\vec{v}\|^2} \quad \text{and} \quad \frac{\vec{u}}{\|\vec{u}\| \|\vec{v}\|} - \frac{\vec{v}}{\|\vec{u}\| \|\vec{v}\|}$$

have the same magnitude where $\vec{u}$ and $\vec{v}$ are nonzero vectors.

66. Figure 13.34 shows that, given three vectors $\vec{u}$, $\vec{v}$, and $\vec{w}$, the sum of the components of $\vec{v}$ and $\vec{w}$ in the direction of $\vec{u}$ is the component of $\vec{v} + \vec{w}$ in the direction of $\vec{u}$. (Although the figure is drawn in two dimensions, this result is also true in three dimensions.) Use this figure to explain why the geometric definition of the dot product satisfies $(\vec{v} + \vec{w}) \cdot \vec{u} = \vec{v} \cdot \vec{u} + \vec{w} \cdot \vec{u}$.

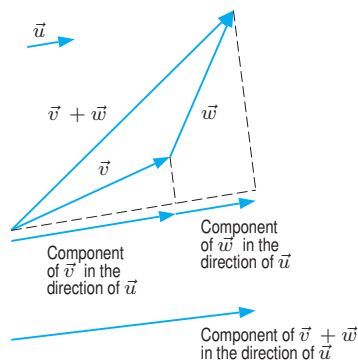


Figure 13.34: Component of $\vec{v} + \vec{w}$ in the direction of $\vec{u}$ is the sum of the components of $\vec{v}$ and $\vec{w}$ in that direction

67. (a) Using the geometric definition of the dot product, show that

$$\vec{u} \cdot (-\vec{v}) = -(\vec{u} \cdot \vec{v}).$$

[Hint: What happens to the angle when you multiply $\vec{v}$ by -1 ?

- (b) Using the geometric definition of the dot product, show that for any negative scalar λ

$$\begin{aligned}\vec{u} \cdot (\lambda \vec{v}) &= \lambda(\vec{u} \cdot \vec{v}) \\ (\lambda \vec{u}) \cdot \vec{v} &= \lambda(\vec{u} \cdot \vec{v}).\end{aligned}$$

68. The Law of Cosines for a triangle with side lengths a , b , and c , and with angle C opposite side c , says

$$c^2 = a^2 + b^2 - 2ab \cos C.$$

On page 735, we used the Law of Cosines to show that the two definitions of the dot product are equivalent. In this problem, use the geometric definition of the dot product and its properties in the box on page 735 to prove the Law of Cosines. [Hint: Let $\vec{u}$ and $\vec{v}$ be the displacement vectors from C to the other two vertices, and express c^2 in terms of $\vec{u}$ and $\vec{v}$.]

69. Use Problems 66 and 67 and the following steps to show (without trigonometry) that the geometric and algebraic definitions of the dot product are equivalent. Let $\vec{u} = u_1\vec{i} + u_2\vec{j} + u_3\vec{k}$ and $\vec{v} = v_1\vec{i} + v_2\vec{j} + v_3\vec{k}$ be any vectors. Write $(\vec{u} \cdot \vec{v})_{\text{geom}}$ for the result of the dot product computed geometrically. Substitute $\vec{u} = u_1\vec{i} + u_2\vec{j} + u_3\vec{k}$ and use Problems 66–67 to expand $(\vec{u} \cdot \vec{v})_{\text{geom}}$. Substitute for $\vec{v}$ and expand. Then calculate the dot products $\vec{i} \cdot \vec{i}$, $\vec{i} \cdot \vec{j}$, etc. geometrically.

70. For any vectors $\vec{v}$ and $\vec{w}$, consider the following function of t :

$$q(t) = (\vec{v} + t\vec{w}) \cdot (\vec{v} + t\vec{w}).$$

- (a) Explain why $q(t) \geq 0$ for all real t .
(b) Expand $q(t)$ as a quadratic polynomial in t using the properties on page 735.
(c) Using the discriminant of the quadratic, show that

$$|\vec{v} \cdot \vec{w}| \leq \|\vec{v}\| \|\vec{w}\|.$$

Strengthen Your Understanding

In Problems 71–73, explain what is wrong with the statement.

71. For any 3-dimensional vectors $\vec{u}$, $\vec{v}$, $\vec{w}$, we have $(\vec{u} \cdot \vec{v}) \cdot \vec{w} = \vec{u} \cdot (\vec{v} \cdot \vec{w})$.
72. If $\vec{u} = \vec{i} + \vec{j}$ and $\vec{v} = 2\vec{i} + \vec{j}$, then the component of $\vec{v}$ parallel to $\vec{u}$ is $\vec{v}_{\text{parallel}} = (\vec{v} \cdot \vec{u})\vec{u} = 3\vec{i} + 3\vec{j}$.
73. A normal vector for the plane $z = 2x + 3y$ is $2\vec{i} + 3\vec{j}$.

In Problems 74–75, give an example of:

74. A point (a, b) such that the displacement vector from $(1, 1)$ to (a, b) is perpendicular to $\vec{i} + 2\vec{j}$.
75. A linear function $f(x, y) = mx + ny + c$ whose graph is perpendicular to $\vec{i} + 2\vec{j} + 3\vec{k}$.

Are the statements in Problems 76–85 true or false? Give reasons for your answer.

76. The quantity $\vec{u} \cdot \vec{v}$ is a vector.

77. The plane $x + 2y - 3z = 5$ has normal vector $\vec{i} + 2\vec{j} - 3\vec{k}$.
78. If $\vec{u} \cdot \vec{v} < 0$ then the angle between $\vec{u}$ and $\vec{v}$ is greater than $\pi/2$.
79. An equation of the plane with normal vector $\vec{i} + \vec{j} + \vec{k}$ containing the point $(1, 2, 3)$ is $z = x + y$.
80. The triangle in 3-space with vertices $(1, 1, 0)$, $(0, 1, 0)$ and $(0, 1, 1)$ has a right angle.
81. The dot product $\vec{v} \cdot \vec{v}$ is never negative.
82. If $\vec{u} \cdot \vec{v} = 0$ then either $\vec{u} = 0$ or $\vec{v} = 0$.
83. If $\vec{u}$, $\vec{v}$ and $\vec{w}$ are all nonzero, and $\vec{u} \cdot \vec{v} = \vec{u} \cdot \vec{w}$, then $\vec{v} = \vec{w}$.
84. For any vectors $\vec{u}$ and $\vec{v}$: $(\vec{u} + \vec{v}) \cdot (\vec{u} - \vec{v}) = \|\vec{u}\|^2 - \|\vec{v}\|^2$.
85. If $\|\vec{u}\| = 1$, then the vector $\vec{v} - (\vec{v} \cdot \vec{u})\vec{u}$ is perpendicular to $\vec{u}$.